

ANDERSON

The Fifteen Watt Tungsten Lamp

Electrical Engineering

M. S.

1912

UNIV OF
ILLINOIS
LIBRARY

ANDERSON

The Fifteen Watt Tungsten Lamp

Electrical Engineering

M. S.

1912

UNIV. OF
ILLINOIS
LIBRARY

The Project Gutenberg eBook of The Fifteen Watt Tungsten Lamp

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Fifteen Watt Tungsten Lamp

Author: Clair Elmore Anderson

Release date: December 2, 2015 [eBook #50588]

Most recently updated: October 22, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/50588

Credits: Produced by Charlene Taylor, Carolyn Jablonski and the
Online Distributed Proofreading Team at
<http://www.pgdp.net>,
in celebration of Distributed Proofreaders' 15th Anniversary
(This file was produced from images generously made
available by The Internet Archive/American Libraries.)

*** START OF THE PROJECT GUTENBERG EBOOK THE FIFTEEN
WATT TUNGSTEN LAMP ***

THE FIFTEEN WATT TUNGSTEN LAMP

BY

CLAIR ELMORE ANDERSON

B. S., University of Illinois, 1911

THESIS

Submitted in Partial Fulfillment of the Requirements for the

Degree of

MASTER OF SCIENCE

IN ELECTRICAL ENGINEERING

IN

THE GRADUATE SCHOOL

OF THE

UNIVERSITY OF ILLINOIS

1912

UNIVERSITY OF ILLINOIS
THE GRADUATE SCHOOL

May 31, 1912

I HEREBY RECOMMEND THAT THE THESIS PREPARED UNDER MY SUPERVISION BY

CLAIR ELMORE ANDERSON

ENTITLED THE FIFTEEN WATT TUNGSTEN LAMP

BE ACCEPTED AS FULFILLING THIS PART OF THE REQUIREMENTS FOR THE

DEGREE OF MASTER OF SCIENCE IN ELECTRICAL ENGINEERING

Emstberg

In Charge of Major Work

Emstberg

Head of Department

Recommendation concurred in:

Morgan Brooks
Elery Blaine
JM Bryant

Committee

on

Final Examination

UNIVERSITY OF ILLINOIS
THE GRADUATE SCHOOL

May 31, 1912

I HEREBY RECOMMEND THAT THE THESIS PREPARED UNDER MY SUPERVISION
BY CLAIR ELMORE ANDERSON ENTITLED THE FIFTEEN WATT TUNGSTEN LAMP
BE ACCEPTED AS FULFILLING THIS PART OF THE REQUIREMENTS FOR THE

DEGREE OF MASTER OF SCIENCE IN ELECTRICAL ENGINEERING

Ernst Berg

In Charge of Major Work

Ernst Berg

Head of Department

Recommendation concurred in:

Morgan Brooks

Ellery B Paine

J M Bryant

Committee on Final Examination

CONTENTS

	Page
I. <u>Introduction,</u>	1
II. <u>Description of Lamps and Tests,</u>	2-4
III. <u>Characteristic Curves,</u>	5-12
IV. <u>Spherical Candle Power,</u>	13- 15
V. <u>Phenomena of “Overshooting”,</u>	16- 18
VI. <u>Theories of “Overshooting”,</u>	19- 20
VII. <u>Amount of “Overshooting”,</u>	21- 22
VIII. <u>Curves of “Overshooting”,</u>	23- 26
IX. <u>Conclusions,</u>	27

THE 15 WATT TUNGSTEN LAMP

I. INTRODUCTION.

Since the introduction of the tungsten lamp some five years ago, the manufacturers have attempted continually to produce smaller and smaller units in the standard voltages. The latest lamp offered today is the 115 volt, 15 watt, tungsten, and it is the purpose of this paper to show the characteristics of this lamp, how it compares with the larger units as to life under different conditions and its behavior in general.

First of all, it must be borne in mind that these tests have been made upon a comparatively small number of lamps, and for that reason the results should not be taken as absolutely conclusive. For the life tests, at least 100 lamps should have been used under each condition, but this was impossible because of the expense.

Special attention has been given to the phenomenon of “overshooting”. An entire year could have easily been spent investigating this subject, and the writer regrets that lack of time has prevented more elaborate and comprehensive tests of this strange phenomenon.

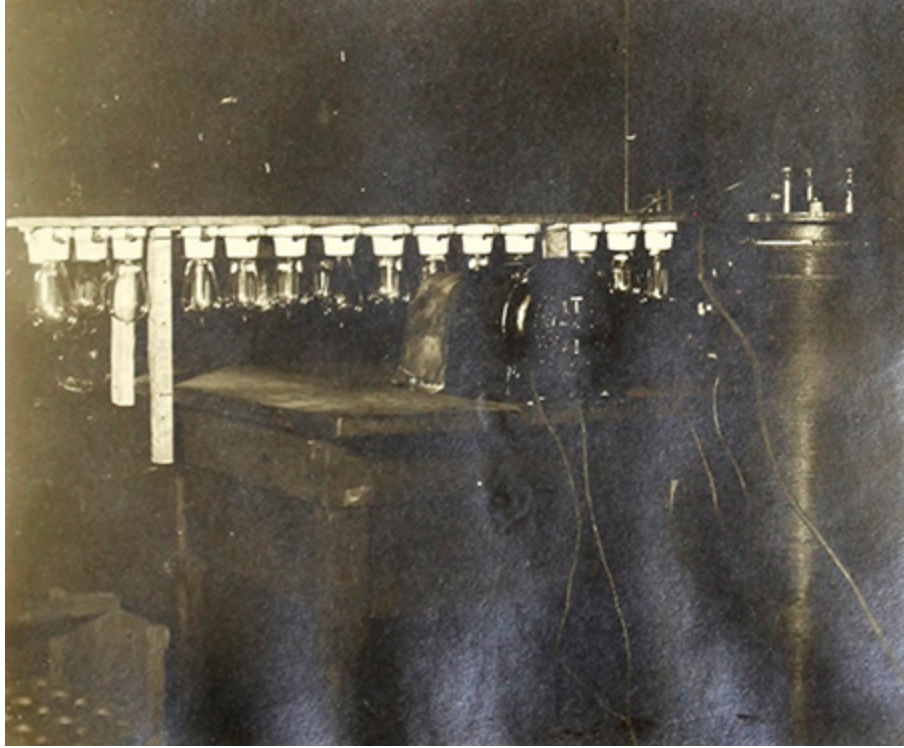
II. DESCRIPTION OF LAMPS AND TESTS.

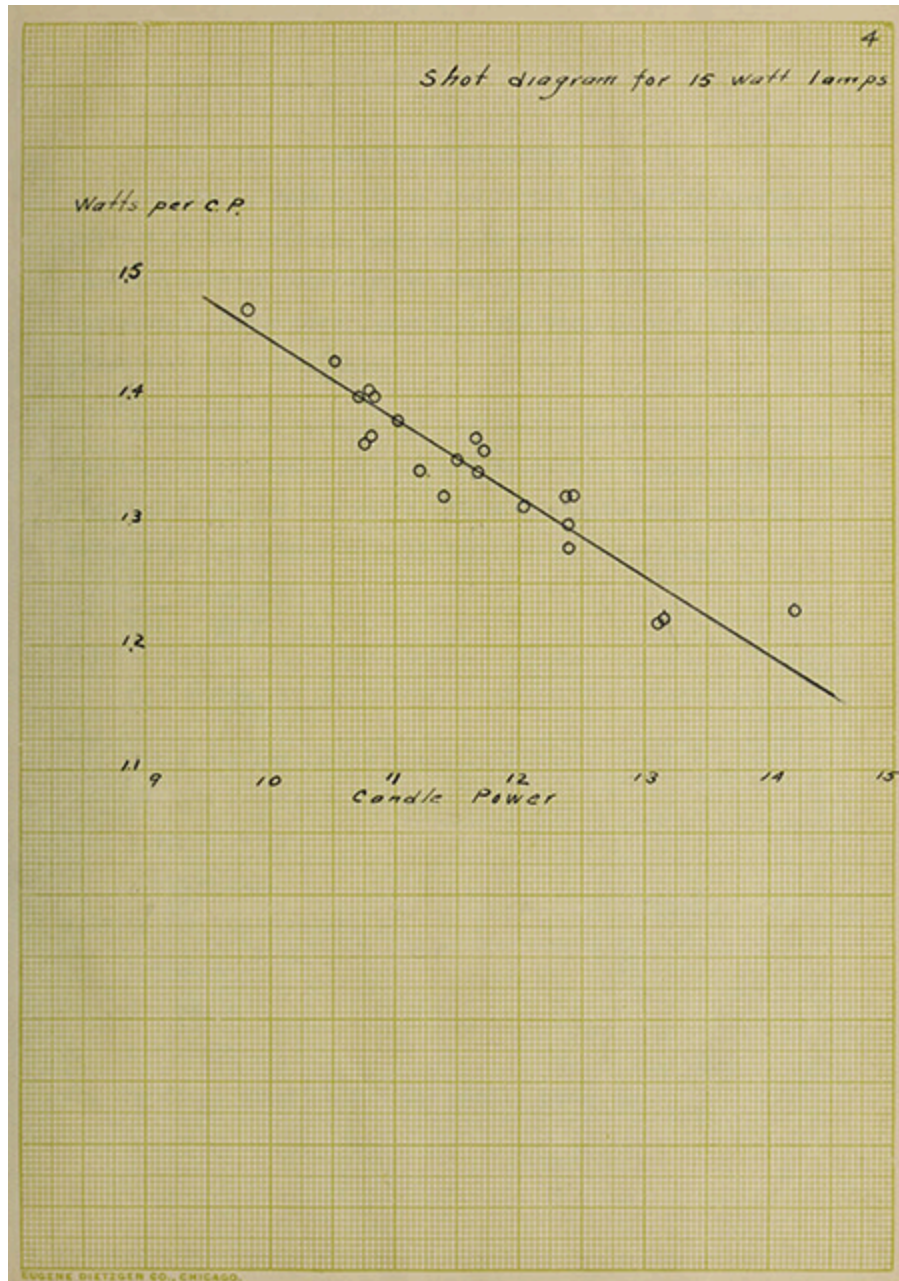
The total number of 15 watt lamps tested was 24, one half of which was obtained directly from the manufacturer and the other half bought in open market. It is well to mention at this time that this may have been the cause of the different qualities as brought out by the life tests.

The lamps were rated at 1.31 watts per horizontal candle power and were supposed to have a useful life of 1000 hours. The voltage ratings of those obtained from the factory were 114 - 112 - 110 and those bought in open market were 115 - 113 - 111. The correct efficiency of the lamps as found by test was 1.34 watts per candle power. A shot diagram follows which shows the actual rating of the lamps at high efficiency.

All readings were made by a Lummer-Brodhun photometer and the voltmeters and ammeters used were carefully standardized. The ammeter was placed beyond the voltmeter in order to get the true current taken by the lamp. The drop across the ammeter was taken into account in the voltmeter readings.

Life tests were made under two conditions, namely, a shock test where the lamps received severe vibrations and a test under ideal conditions, i.e. no jar and constant voltage. In order to obtain vibrations for the lamps upon the shock test, a small motor, with its shaft pulley off set, was screwed rigidly to a table. The lamps were placed in a normal position upon the table by means of wooden frames. The result was that when the motor was running it had a pounding effect, thus putting the table, consequently the lamps, in a state of severe vibration. The filaments of the lamps could be seen violently shaking for some distance. The test was indeed a hard one, and one that would not be found in many actual cases. It is very doubtful if railway lamps are subjected to such a strain and they are of the heavy filament low voltage type. The following photograph shows the arrangement above described. Ten 15 watt lamps were used on this test, the remainder shown being 20 and 25 watt and carbons.





Shot diagram for 15 watt lamps

III. CHARACTERISTIC CURVES.

Figure 1, Page 8, shows the variations of the candle power with the voltage, current and watts. Figure II shows the relation between candle power and the efficiency, watts per horizontal candle power, and also the variation of the candle power with the resistance.

An empirical formula for the candle power expressed as a function of the watts is $cp = KW^x$ where K is a constant of the lamp and W denotes the watts. From the curve when $cp = 5$, watts = 11.1 and when $cp = 15$, watts = 17.5 dividing

$$cp_1/cp_a = KW_1^x/KW_a^x$$

substituting

$$5/15 = 11.1^x/17.5^x$$

and

$$\begin{aligned}\log 5 + x \log 11.1 &= x \log 17.5 \\ .4771 + 1.0453x &= 1.2430x \\ .198x &= .4771 \\ x &= 2.41\end{aligned}$$

solving for the constant K

$$\begin{aligned}5 &= K 11.1^{2.41} \\ 5 &= 332 K \\ K &= .0150\end{aligned}$$

and the final equation for the candle power is

$$cp = .0150 \times w^{2.41}$$

In the same way, the candle power may be expressed in terms of the voltage and this is found to be

$$cp = 334 \times 10^{-9} E^{3.68}$$

This formula checks precisely with the one used in the engineering department of the General Electric Company at their lamp works, Harrison, N.J.

The horizontal distribution curve of a lamp with its filament mounted as is the modern tungsten is nearly a circle. This is not true, however, in the case of vertical distribution and this curve is shown, Figure [III](#). As will be noted, the tip candle power is only about 23 per cent of the horizontal.

Life Tests.

The results of the life tests were very surprising. The lamps upon the test under [ideal](#) conditions, namely, no vibrations and constant voltage, had only an average life of 460 hours, while every one of those upon the [shock](#) test are still burning at the present time, having been burned 300 hours. In order to make the test still more severe, the lamps were subjected to vibrations without voltage being impressed, and as yet, not a filament has broken, the total time being 400 hours. It was impossible to give more time to these lamps as was done for those under ideal conditions, for the reason it was thought inadvisable to leave the motor, which gave the vibrations, running over night.

The curves have the same general form for the two conditions but the variations are far more great for the lamps which were upon the shock test. The reason for this is that the vibrations were so severe as to shake parts of the filament together thus giving a partial short circuit, causing great variations in candle power.

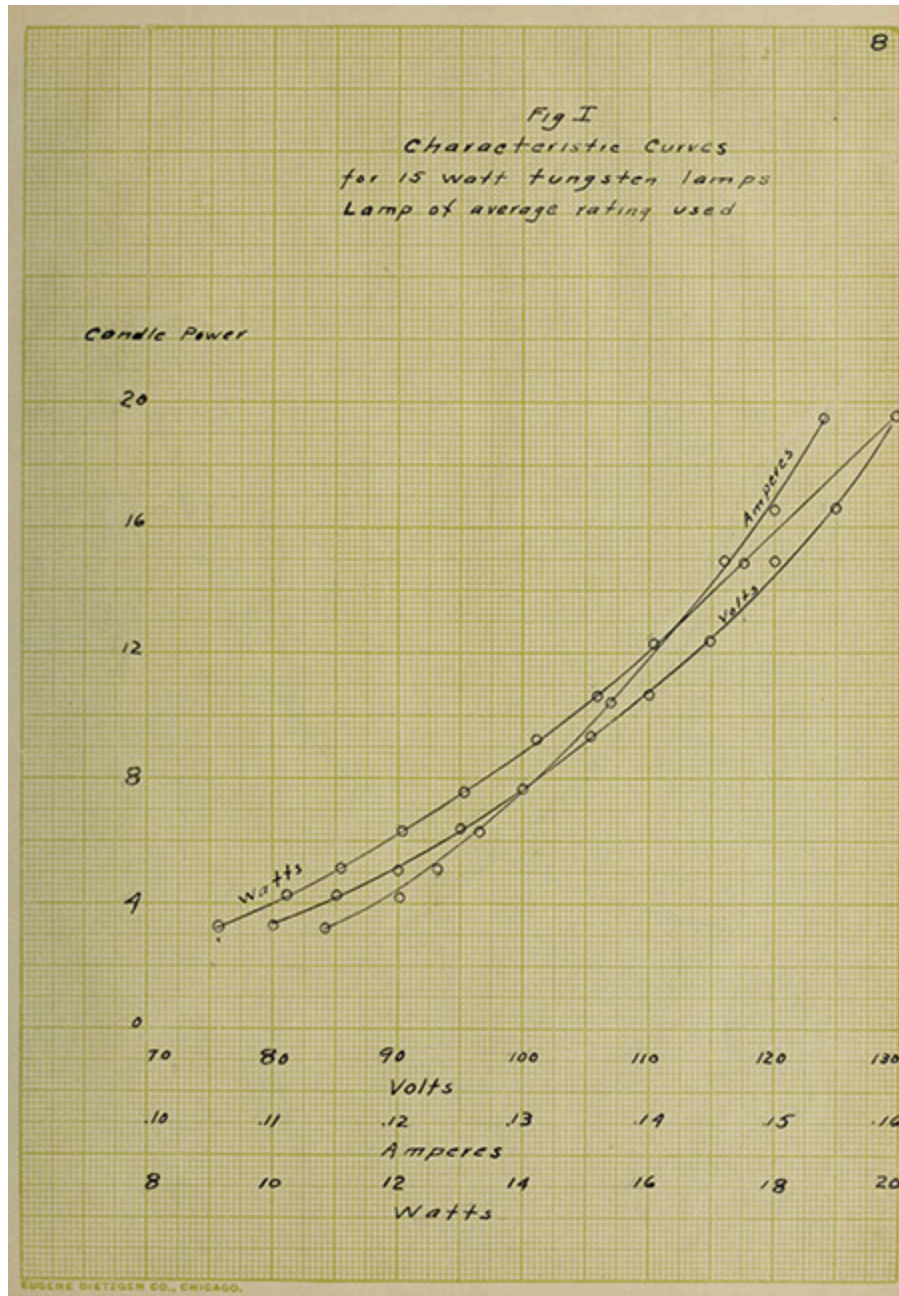
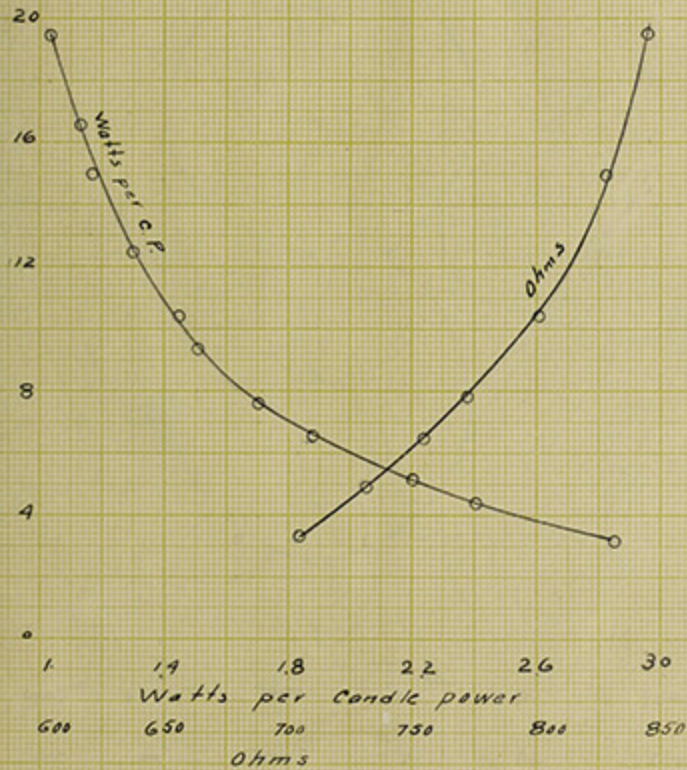


Fig I

Characteristic Curves for 15 watt tungsten lamps
Lamp of average rating used

Fig II
Characteristic Curves
for 15 watt tungsten lamps
Lamp of average rating used

Candle Power



EUGENE DIETZGEN CO., CHICAGO.

Fig II

Characteristic Curves for 15 watt tungsten lamps

Lamp of average rating used

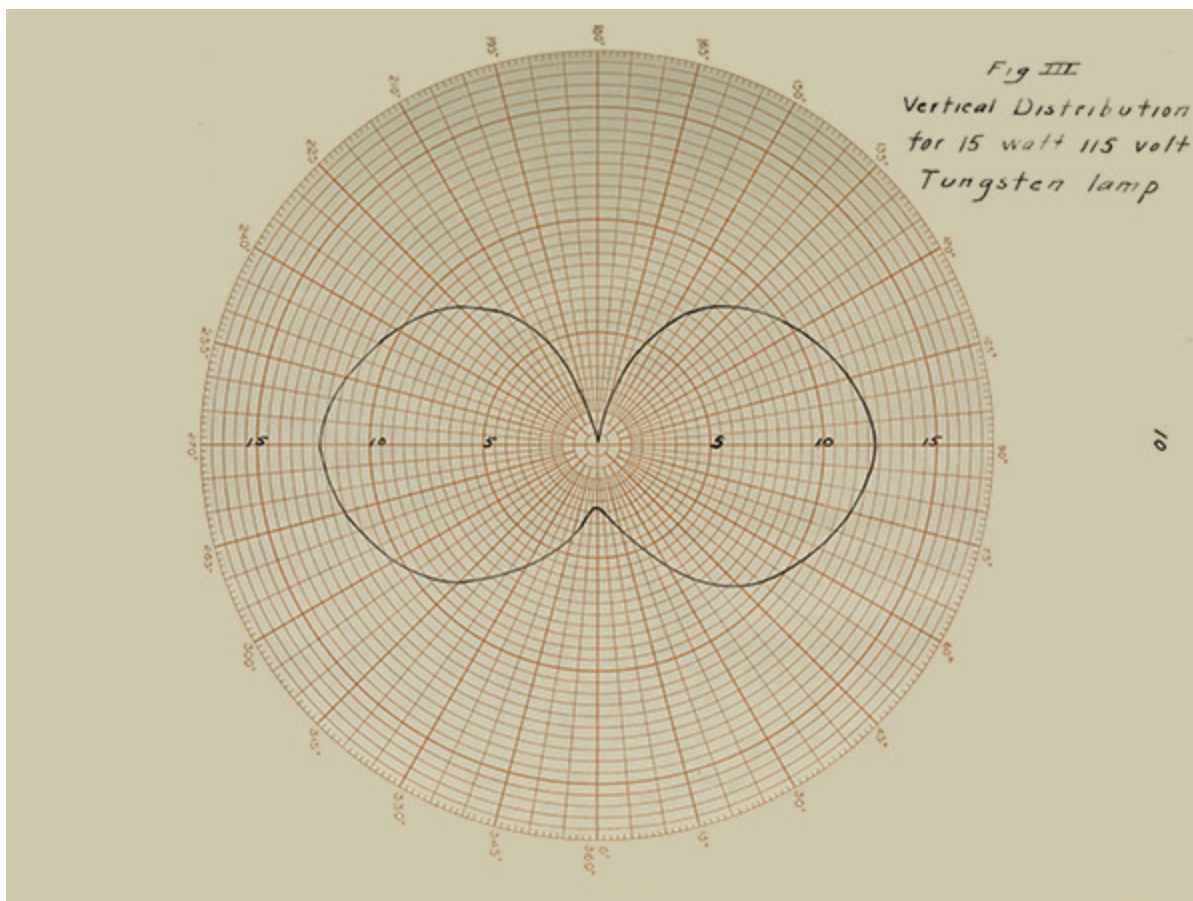
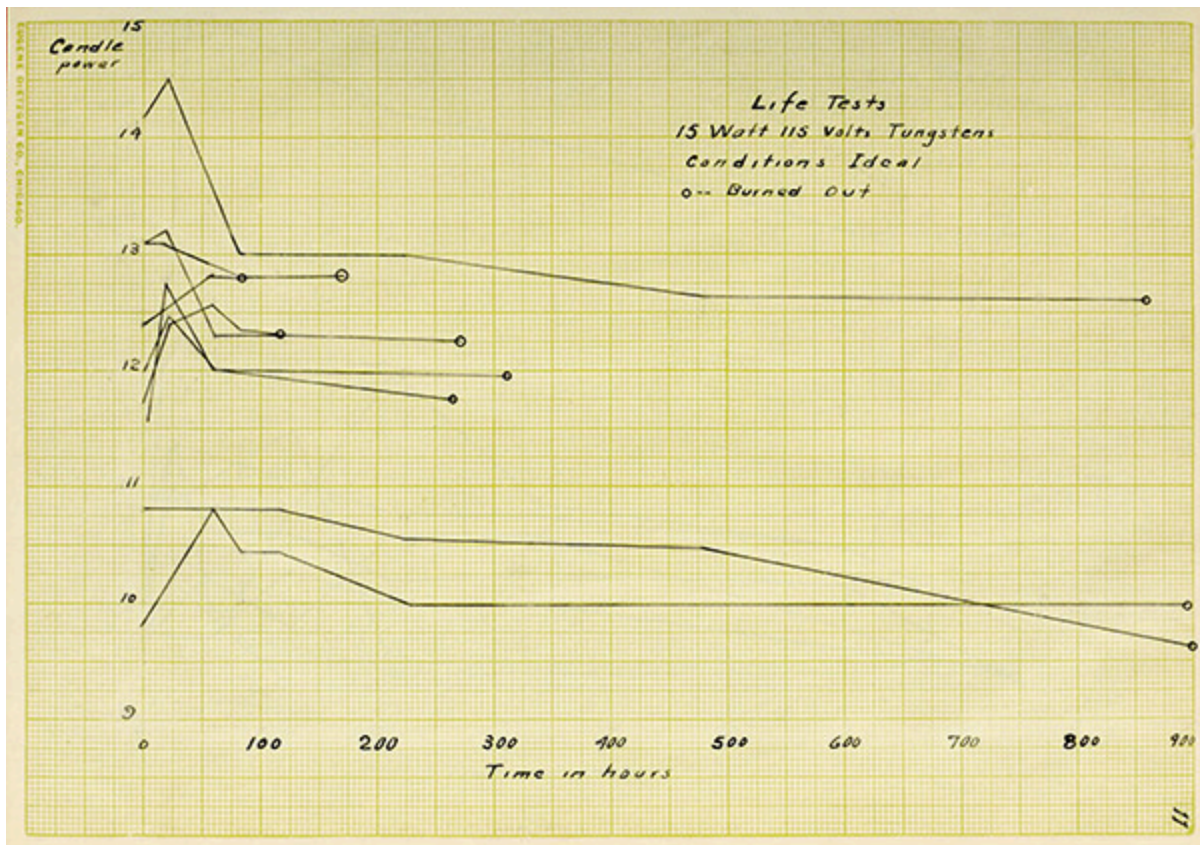
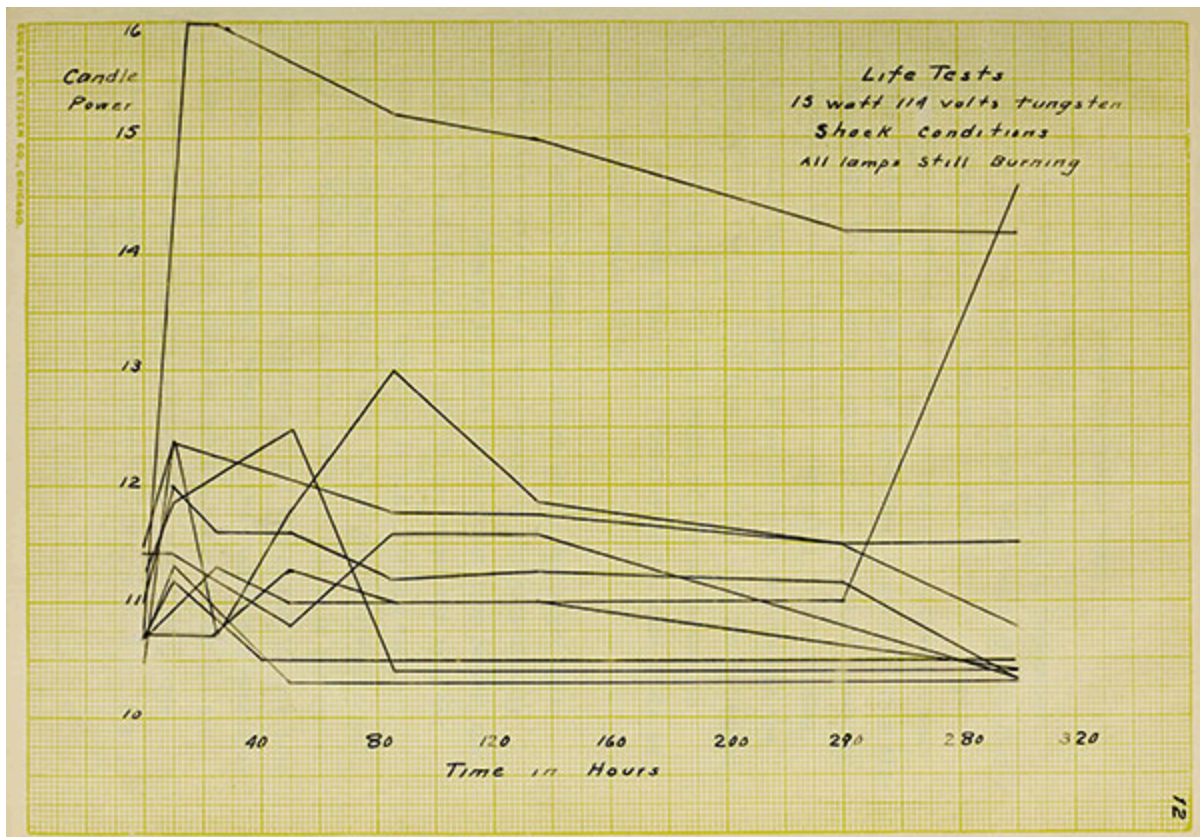


Fig III

Vertical Distribution for 15 watt 115 volt Tungsten lamp



Life Tests
 15 Watt 115 Volts Tungstens
 Conditions Ideal
 o—Burned Out



Life Tests
 15 watt 114 volts tungsten
 Shock Conditions
 All lamps still Burning

IV. SPHERICAL CANDLE POWER.

Owing to the absence of an integrating photometer, the mean spherical candle power has been found by Kennelley's graphical method. This method is very simple as compared with Rousseau's and has the advantage of yielding the mean spherical intensity as a one dimensional quantity. This dispenses with the use of a planimeter or equivalent measuring surface device. It consists essentially in determining graphically from the given polar curve an evolute and the involute of the same and then projecting this involute upon a vertical line. Half the length of the projection is equal to the mean spherical intensity to the same scale as the original polar curve.

Figure [IV](#) shows the method and is explained thus: The polar curve O A H B corresponds to the distribution of intensity from an inverted incandescent lamp having its base at V and tip at V'. This is not precisely true but the only variation to speak of is that the tip candle power does not fall off so much as the curve between B and O. This variation is slight, however. The mean horizontal intensity is OH, the diameter of the circle, and in this case is equal to 12.4 candle power. In the diagram the construction is adapted to zones of 30° and the radii of the midzones found i.e. at $+75^\circ$, $+45^\circ$, $+15^\circ$, -15° , -45° -75° . These are marked by dotted lines O t, O s, O r, O r', O s', O t' respectively.

With radius O r and center O, the arc hra is described through an angle of 30° . The radius Oa is drawn at the end of the arc. A distance Ab is measured from a along a O equal to O s, the second midzone radius. With a center b and radius O s, the arc ac is described through an angle of 30° so that bc makes an angle of 60° with the horizontal OH. The line bc is drawn at the end of this arc. From c towards b, a distance cd is marked off equal to O t, the third midzone radius. With center d and radius O t, the arc ce is described through an angle of 30° so that de makes an angle of 90° with the horizontal OH. The line de is drawn.

The arc $ha'c'e'$ is extended from the horizontal to the vertical beneath in the same manner as above by steps of 30° with centers O, b' , and d' , and radii Or' , OS' and Ot' respectively. The curve $ecarr'a'c'e'$ is now continuous and complete. A vertical line QQ' is drawn through the convenient point H and the points $e c a a' c' e'$ are projected upon the same. The length HQ, is the upper hemispherical intensity and the length HQ' the lower hemispherical intensity. Their arithmetical mean is the mean spherical intensity. Since in this case the upper and lower hemispheres are symmetrical $HQ = HQ' = QQ'/2 = \text{mean spherical intensity}$. By measurement this half length is found to be 3.125 inches and from the scale used this corresponds to 9.67 candle power. The spherical reduction factor for these lamps is, then

$$9.67/12.4 = 78\%$$

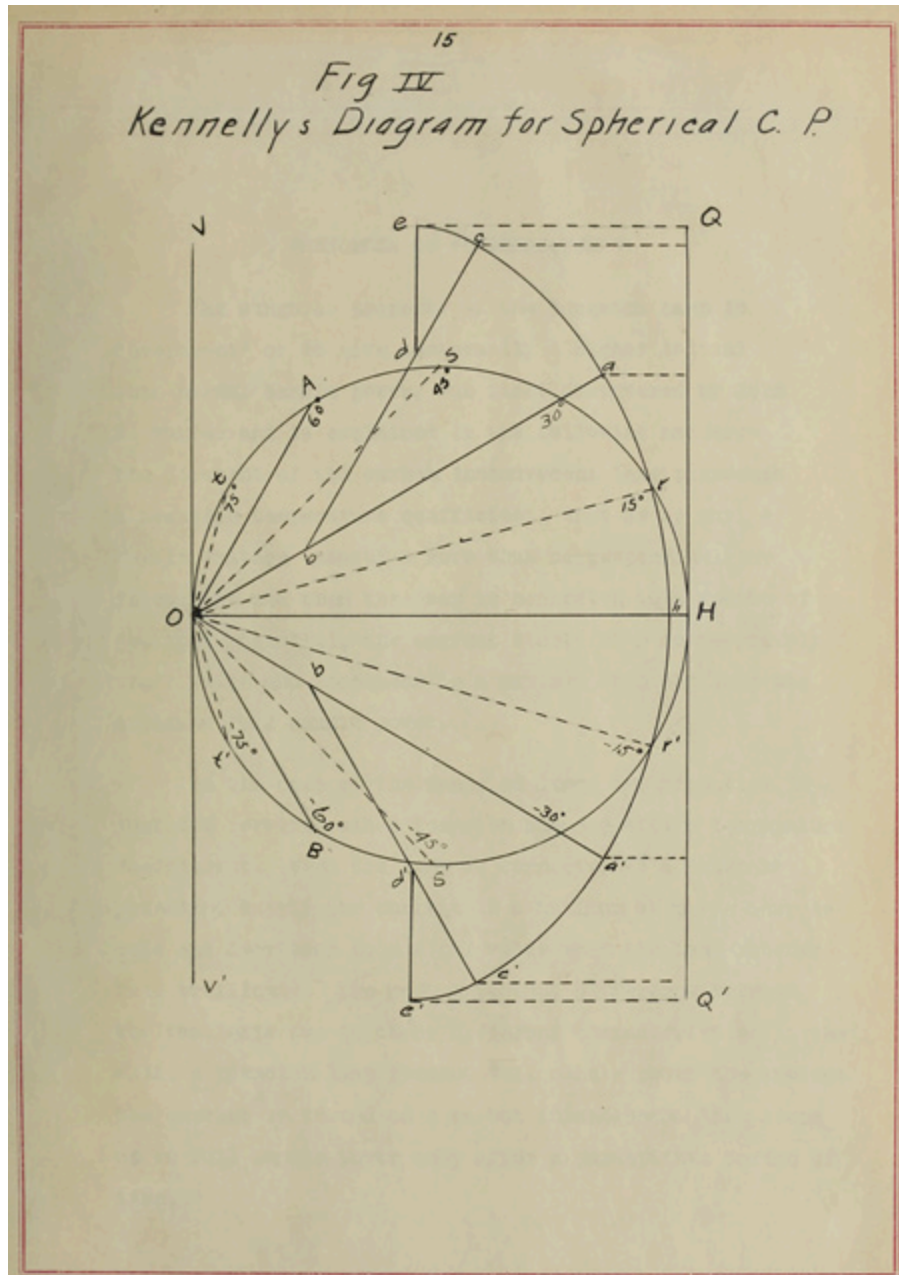


Fig IV
Kennelly's Diagram for Spherical C. P.

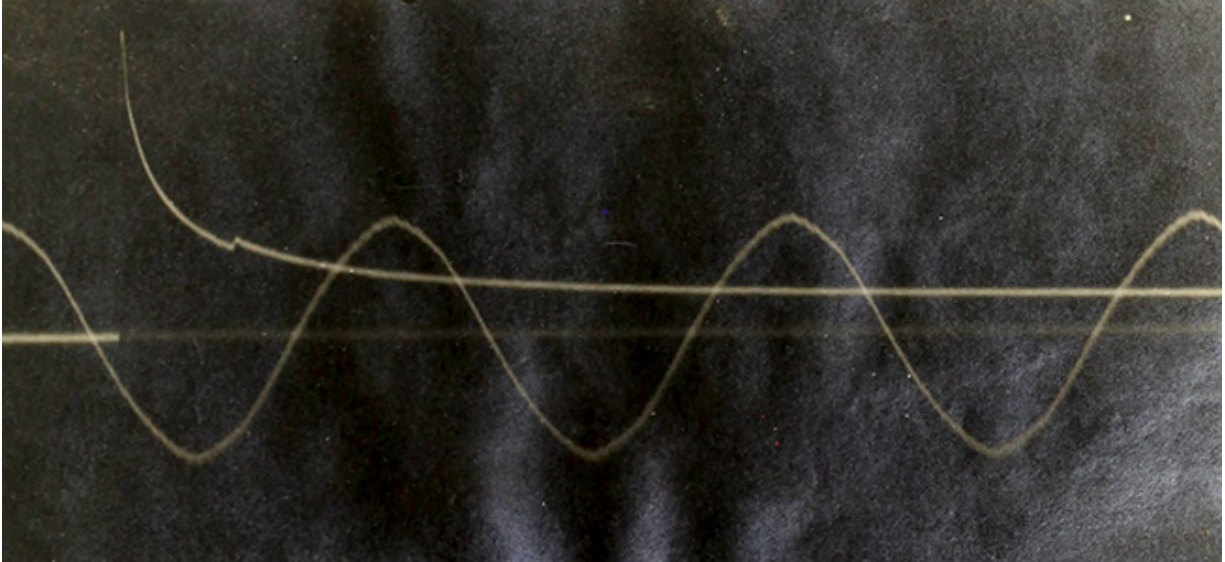
V. PHENOMENA OF “OVERSHOOTING”.

The singular property of the tungsten lamp to “overshoot” or to give temporarily a higher initial than normal candle power, was first discovered by John B. Taylor and is explained in the following manner:—The filament of the carbon incandescent lamp possesses a negative temperature coefficient; that is to say, a rise in voltage causes a more than corresponding rise in current and when the lamp is connected to a source of constant potential, the current starts at a comparatively small value and increases to a maximum when the lamp has attained full candle power.

In the case of the tungsten lamp, the situation is just the reverse, since tungsten has a positive temperature coefficient. When the lamp is connected to a constant potential supply the current is a maximum when the lamp is cold and decreases to a final value when the lamp reaches full brilliancy. The most important difference between the two lamps due to these different characteristics is that while a tungsten lamp reaches full candle power the instant the current is turned on a carbon incandescent lamp comes up to full candle power only after a perceptible period of time.

The apparent temporary increase in the candle power of a tungsten lamp was observed early after the lamp was invented but it was generally ascribed to some possible physiological action due to the slow contraction of the pupil of the eye.

The following curve was obtained by means of the oscillograph and shows clearly the rush of current for the first instant after the lamp is turned on. The break in the curve is due to an imperfection in one of the operating switches and has nothing to do with any characteristic of the lamp. The cycle wave was put on merely to obtain the time.



In order to prove that this overshooting occurs, an actual photograph of the intensity has been made. This was obtained by making a box $1' \times 1' \times 3'$ absolutely light proof and arranging a lamp inside so that it could be turned on and off at will. A slit, fitted with a shutter, was cut in one end of the box which permitted the light to fall upon a revolving oscillograph film. The film holder was attached to the box with thin metal strips and revolved by means of a small motor. As is seen the whole arrangement was nothing more than a large camera.

The following photograph shows the phenomenon quite clearly, point A denoting where the lamp was turned on.



VI. THEORIES OF “OVERSHOOTING.”

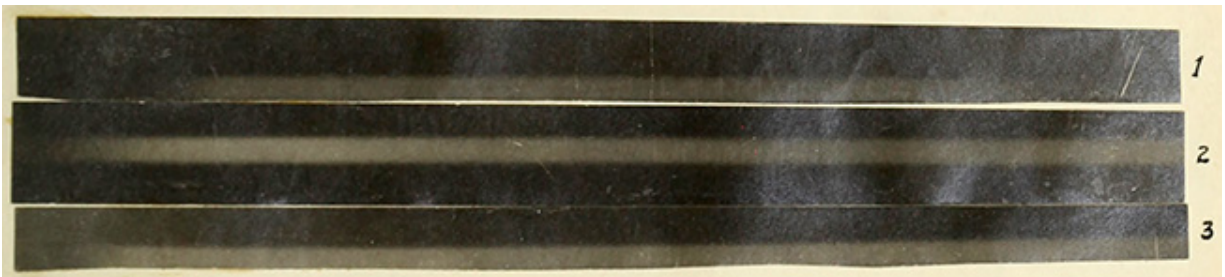
The theory given by Taylor to account for the “overshooting” of tungsten lamps is based on the fact that there is a small amount of residual gas in the lamp, which is attracted to the walls of the lamp when it is cold; and when the lamp is lighted and warms up, this residual gas is driven off lowering the vacuum. With a high vacuum, practically all the energy must be radiated from the filament; conversely, on a lowering of the vacuum, some of the heat is carried away by convection and conduction. When all the heat is carried away by radiation the filament runs at a higher temperature and will give more light.

Another theory is that a cold tungsten filament lamp absorbs and occludes certain gaseous substances from the low pressure space within the chamber. Owing to the presence of these gases the filament shines more brightly when first brought quickly to incandescence, but after the gases have been driven off by the heat, the extra luminescence disappears and can be regained only by prolonged cooling and rest.

Still another theory, and the one that seems the most logical to the writer, is that the increase of resistance accompanying the rise of temperature takes a certain small interval of time so that when the temperature is rising at the rate of thousands of degrees per second, the resistance lags perceptibly. The resistance does not suppress the current as quickly as it should and an extra rush of current and heat energy goes through the filament, raising the temperature above normal, with a corresponding increase in brilliancy.

VII. AMOUNT OF "OVERSHOOTING".

In order to determine the amount of over shooting, the writer has made photographs as shown below. Number one was made with the lamp under voltage, number two by using normal voltage and suddenly turning the lamp on by means of a snap switch, thereby obtaining the overshooting, and finally number three was made by impressing voltage above normal. The pictures were obtained by using the photographic arrangement as before described. All three prints were made from the same film, that is the three pictures were made upon one film thus insuring the same development and printing for all. The print has been cut merely to allow a closer comparison of the intensities.



It is seen that number three compares favorably with the overshooting as shown by number two and the candle power corresponding was found to be approximately 50% greater than normal. It is not claimed that every lamp will overshoot this amount as the degree of vacuum or other factors of individual lamps may play an important part in this phenomenon. There is no doubt, however, that this strange fact really occurs and is not due to physiological reasons.

VIII. CURVES OF “OVERSHOOTING”.

In order to prove that the law of resistances, namely, $R = R_0(1 + \alpha t)$ does not hold for the first instance after closing the switch on a tungsten lamp, the following curves have been plotted. Number [V](#) has been taken from the oscillograph record shown in the first part of this paper and shows that about .024 second elapses before the current becomes normal. Knowing the current at any instant as given by this curve, it is easy to find the resistance at the same instant by Ohm's law, the electromotive force being a constant and known value. Curve [VI](#) shows this relation. Curves [VII](#) and [VIII](#) are approximate values and not absolute. Now from the temperature curve, values are taken and substituted in the formula for resistance, $R = R_0(1 + \alpha t)$, the resulting curve being Figure [IX](#). It is seen that curves [VI](#) and [IX](#) do not take the same values at all until after a brief interval of time has elapsed. Curve [VI](#) is absolutely correct, however, as these values have been obtained from the oscillograph record. Consequently, the assumption upon which curve [IX](#) is based must be incorrect for the first .024th of a second and the conclusion is that the law of resistances does not hold. This result tends to strengthen the theory of the lag of resistivity for the “overshooting” of a tungsten lamp.

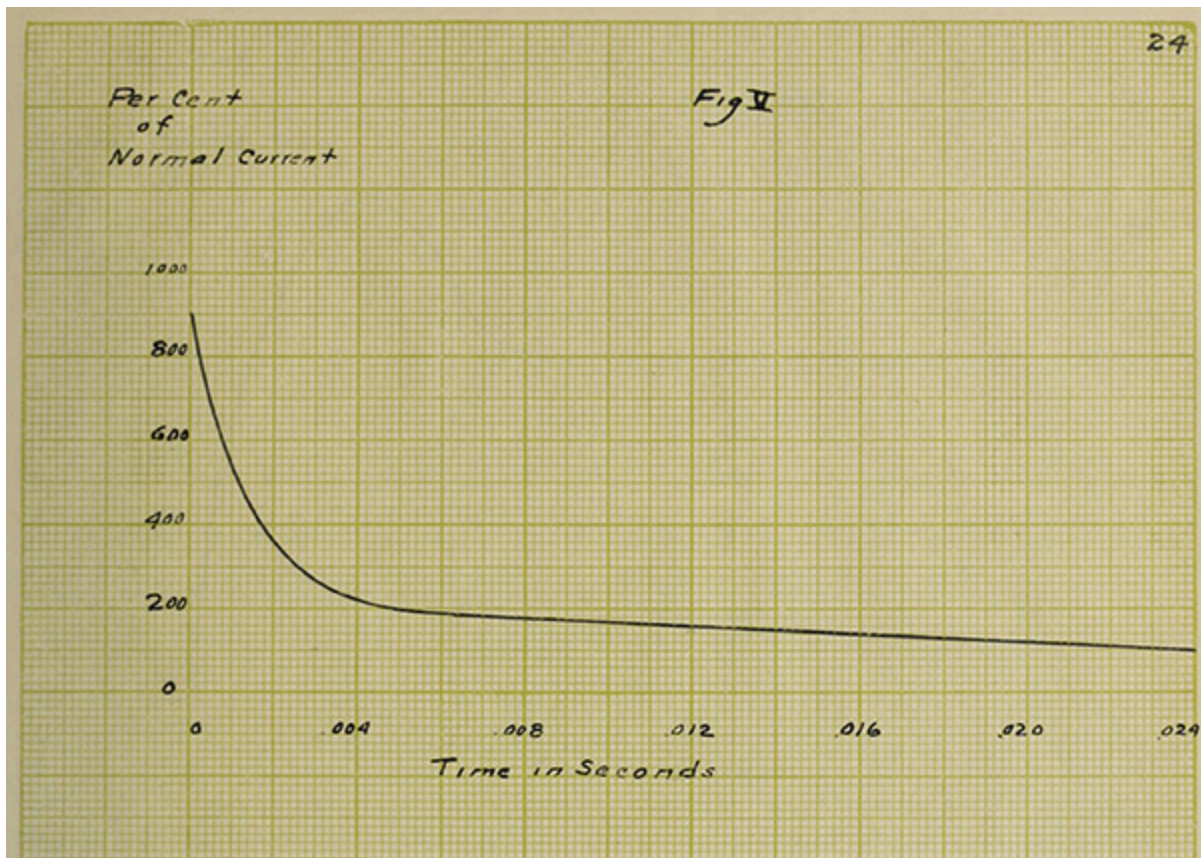


Fig V

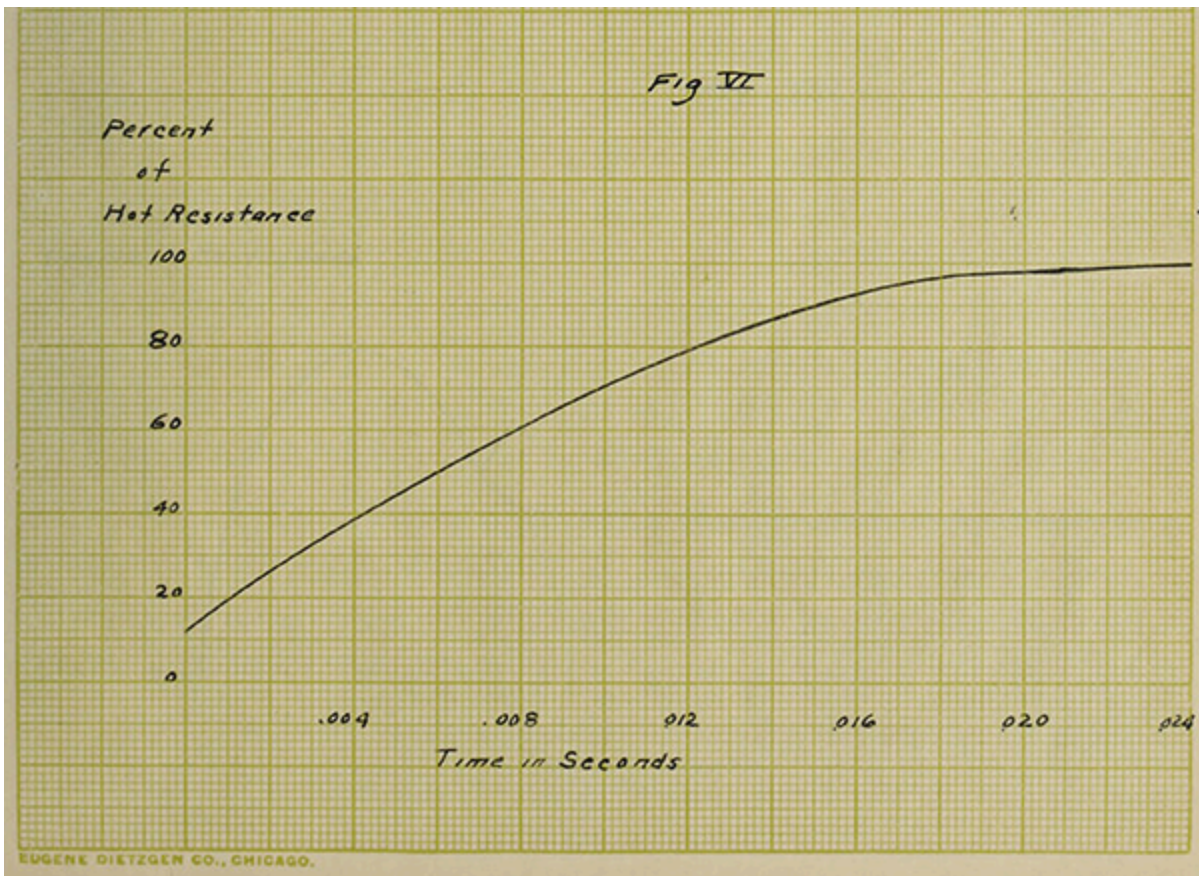


Fig VI

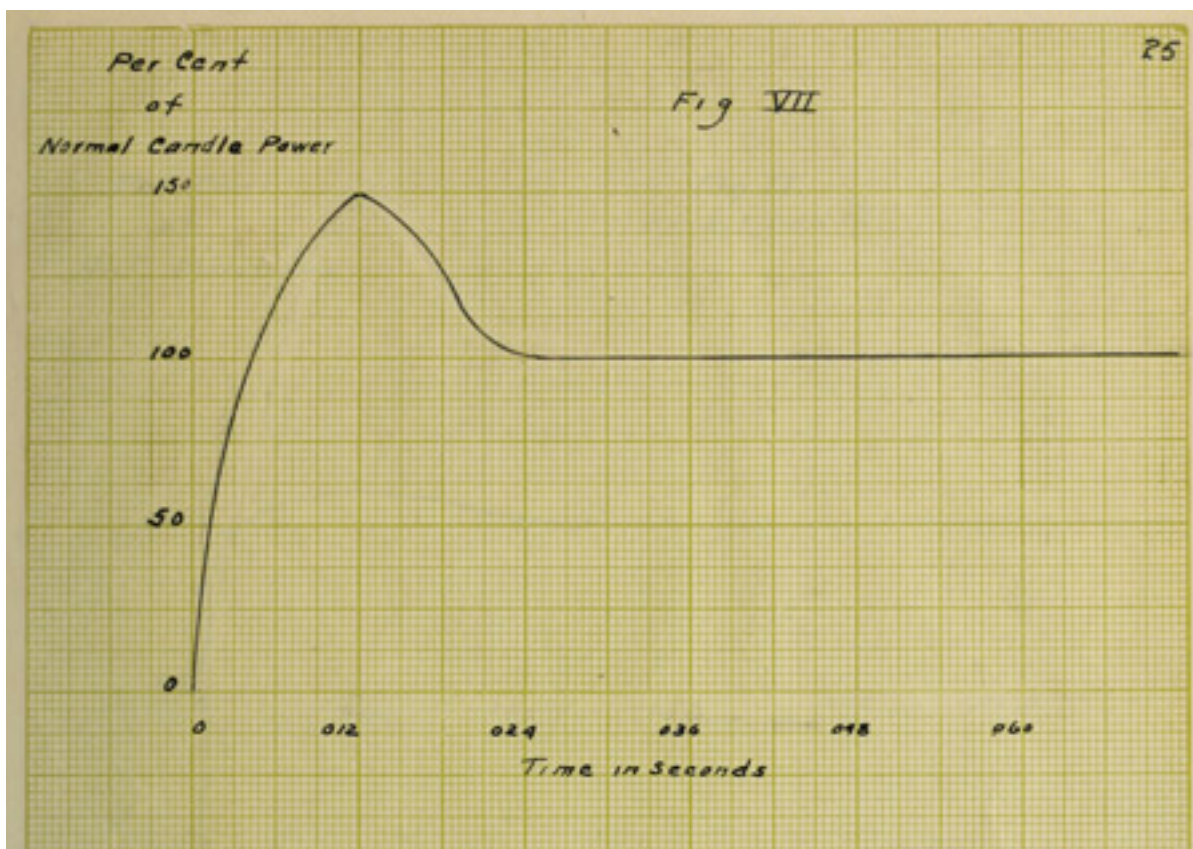


Fig VII

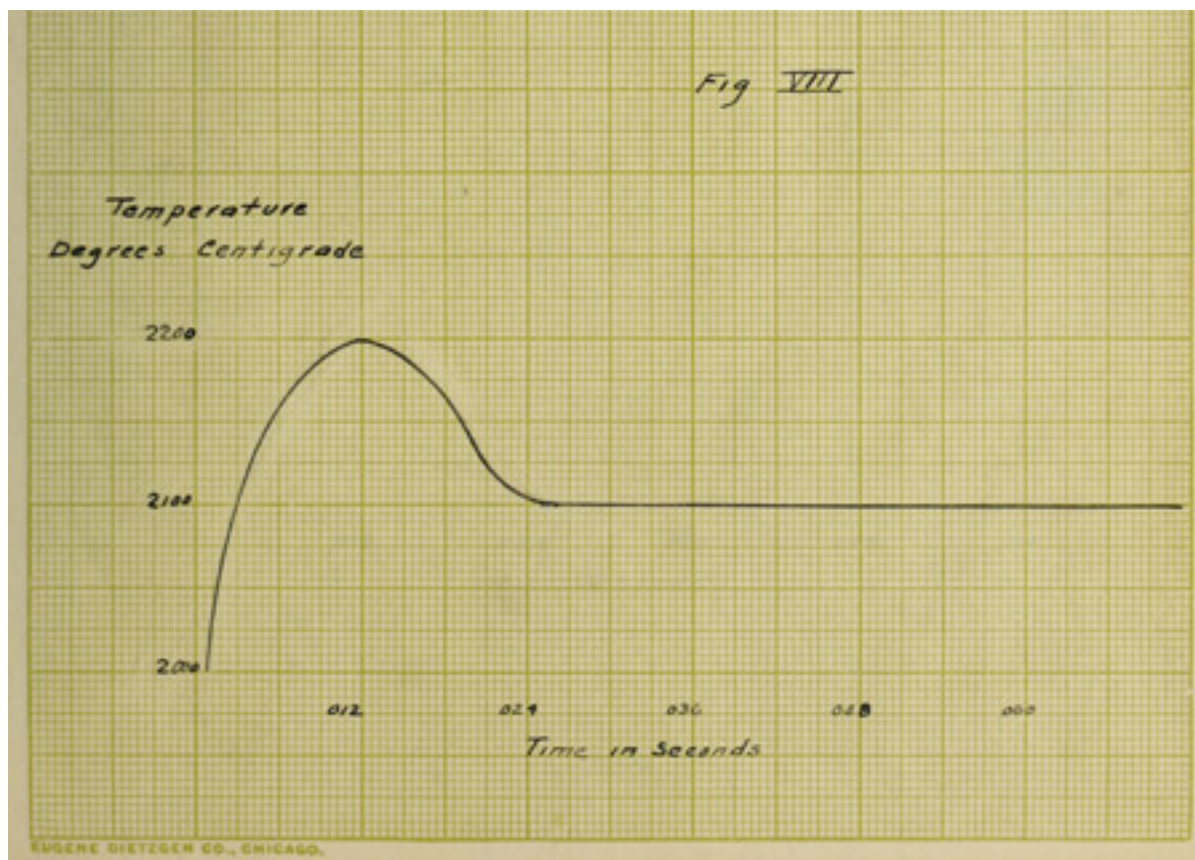


Fig VIII

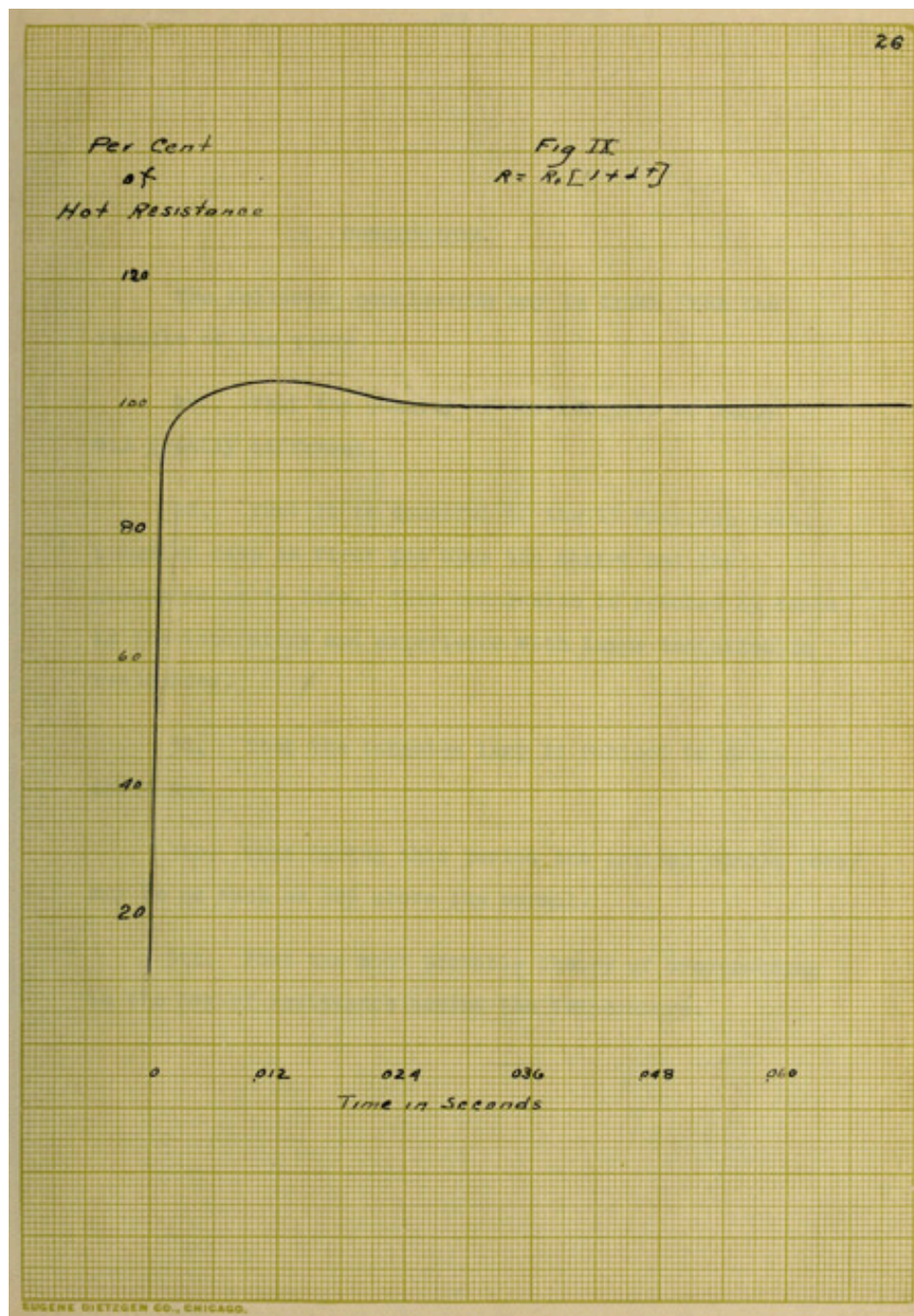


Fig IX

$$R = R_0 [1 + \alpha t]$$

IX. CONCLUSIONS.

The following conclusions may be drawn from the results of the tests.

1st, That the quality of the two sets of lamps was greatly different.

2d, That it is doubtful if the 15 watt, 115 volt, tungsten lamp as first put upon the market met the guarantee as to life. This conclusion is reached by tests in the laboratory and experience with lamps installed in residences.

3d, That the tungsten lamp is subject to overshooting.

4th, That during this period the initial candle power may be as much as 50% above the normal.

5th, That the most probable theory of overshooting is the lag of resistance behind the temperature.

TRANSCRIBER'S NOTE

Fractions are represented horizontally using the slash / symbol. The original separated the numerator and denominator horizontally, separated by a _ line.

Punctuation has been normalized. Except in the following cases, variations in hyphenation, spelling and spacing have been retained as they were in the original publication.

[Chapter 2:](#)

Lumer-Brodhum photometer —> Lummer-Brodhun photometer

[Chapter 4:](#)

What appear to be hand-drawn primes in the text have been confirmed as: The arc ha'ce —> The arc ha'c'e'.

e c a a' c e' —> e c a a' c' e'

[Figure IV](#)

There are two points b radiating from point O. The lower is likely to represent b'.

[Chapter 5:](#)

inportant difference between —> important difference between

*** END OF THE PROJECT GUTENBERG EBOOK THE FIFTEEN
WATT TUNGSTEN LAMP ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.